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**Department of Mathematics**
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# Math 208 Final Exam Term 253

Original exam date: 5 August 2026. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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**1.** Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$, which has an eigenvalue $\lambda = 2$ of multiplicity 3 and defect 2. If we choose the generalized eigenvector $v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ such that $(A-2I)^3 v_3 = 0$ but $(A-2I)^2 v_3 \neq 0$, which of the following represents the general solution of the differential system $X' = AX$?

(a) $X(t) = \left( c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ t \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ t \\ t^2/2 \end{bmatrix} \right) e^{2t}$

(b) $X(t) = \left( c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ t \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} t \\ t^2/2 \\ 1 \end{bmatrix} \right) e^{2t}$

(c) $X(t) = \left( c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ t \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} \right) e^{2t}$

(d) $X(t) = \left( c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t+1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} t^2 \\ 2t \\ 1 \end{bmatrix} \right) e^{2t}$

(e) $X(t) = \left( c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} t^2/2 \\ t \\ 1 \end{bmatrix} \right) e^{2t}$

**2.** If the rank of the matrix $A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & -3 & 5 & 4 \\ 3 & -5 & 8 & 5 \\ 1 & -1 & 2 & k^2 - 6 \end{bmatrix}$ is equal to 2, then $k =$

(a) $\pm 3$

(b) $\pm\sqrt{6}$

(c) $-3$ only

(d) 0

(e) 3 only

**3.** If $K = \begin{bmatrix} \alpha \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector with eigenvalue $\lambda = -2$ of the matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ -3 & 1 & -3 \\ 1 & -4 & 2 \end{bmatrix}$, then $\alpha =$

(a) 2

(b) 1

(c) $-2$

(d) 0

(e) $-1$

**4.** Consider the non-homogeneous system

$$X' = AX + \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

If the general solution of the associated homogeneous system is $X_c = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t}$, then a particular solution $X_p$ is

(a) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

(e) $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$

**5.** The solution of the exact differential equation

$$\left(\sin y + y \sin x + \frac{1}{x}\right) dx + \left(\frac{1}{y} + x \cos y - \cos x\right) dy = 0$$

is given by

(a) $\ln|x| - \ln|y| - x \cos y - y \cos x = C$

(b) $\ln|x| + \ln|y| + x \sin y - y \cos x = C$

(c) $\ln|x| + \ln|y| + x \cos y - y \cos x = C$

(d) $\ln|x| + \ln|y| + y \sin y - x \cos x = C$

(e) $\ln|x| - \ln|y| + x \sin y - y \cos x = C$

**6.** The minimum radius of convergence of a power-series solution of the differential equation

$$(x^2 - 4x + 13)y'' + 2xy' - y = 0$$

about the ordinary point $x = 1$ is equal to

(a) $\sqrt{13}$

(b) 3

(c) $\sqrt{10}$

(d) $2\sqrt{13}$

(e) 5

**7.** Consider the system

$$X' = AX, \qquad X(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

where $A$ is a $2 \times 2$ matrix with real entries. If $A$ has the eigenvalue $\lambda = 1 + i$ with corresponding eigenvector $K = \begin{bmatrix} -1 - i \\ 1 \end{bmatrix}$, then $X(\pi) =$

(a) $e^{\pi} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

(b) $e^{\pi} \begin{bmatrix} -2 \\ -1 \end{bmatrix}$

(c) $e^{\pi} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(d) $e^{\pi} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(e) $e^{\pi} \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

**8.** The sum of the indicial roots at $x = 0$ for the differential equation

$$x(4 - x)y'' + (2 - x)y' + 4y = 0$$

is

(a) 0

(b) $-\frac{1}{4}$

(c) $-\frac{1}{2}$

(d) $\frac{1}{2}$

(e) $\frac{1}{4}$

**9.** A fundamental matrix of the system $X' = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} X$ is

(a) $\begin{bmatrix} 2e^{-t} & e^{3t} \\ -e^{-t} & e^{3t} \end{bmatrix}$

(b) $\begin{bmatrix} -e^{-t} & e^{3t} \\ e^{-t} & e^{3t} \end{bmatrix}$

(c) $\begin{bmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{bmatrix}$

(d) $\begin{bmatrix} e^{-t} & 3e^{3t} \\ -e^{-t} & e^{3t} \end{bmatrix}$

(e) $\begin{bmatrix} e^{-t} & e^{3t} \\ e^{-t} & e^{3t} \end{bmatrix}$

**10.** The general solution of $X' = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} X$ can be written as

$$X = c_1 \begin{bmatrix} 1 \\ \alpha \\ 1 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 0 \\ \beta \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} \gamma \\ -2 \\ 1 \end{bmatrix} e^{t}.$$

Then $\alpha + \beta + \gamma =$

(a) $-1$

(b) 3

(c) 0

(d) $-2$

(e) 1

**11.** Suppose that a $2 \times 2$ matrix $A$ has the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 1$, with corresponding eigenvectors $v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, respectively. If $A^2 = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$, then $x + y + z + w =$ (Hint: Find a diagonalizing matrix $P$ and write $A = PDP^{-1}$.)

(a) $-9$

(b) $-7$

(c) $-6$

(d) $-10$

(e) $-5$

**12.** A general solution of the differential equation $\dfrac{dy}{dx} + \dfrac{2}{x}y = 4x^3y^4$ is given by

(a) $y^{-3} = 6x^4 + Cx^6$

(b) $y^{-3} = 4x^4 + Cx^6$

(c) $y^{-3} = 2x^4 + Cx^6$

(d) $y^{-3} = 3x^4 + Cx^6$

(e) $y^{-3} = 5x^4 + Cx^6$

**13.** The number of irregular singular points of the differential equation

$$x^2(1 - x^2)y'' + y' + xy = 0$$

is

(a) 1

(b) 4

(c) 2

(d) 3

(e) 0

**14.** By using the method of undetermined coefficients, a particular solution of the differential equation

$$y'' - 2y' = 6e^{2x} + 5\cos x$$

is given by $y_p = Axe^{2x} + B\cos x + C\sin x$. Then $5B + 2A + 5C =$

(a) $-9$

(b) $-3$

(c) 0

(d) 7

(e) 5

**15.** A particular solution of the differential equation $y'' + 4y = 4\sec(2x)$ is given by $y_p(x) =$

(a) $\sin(2x)\ln|\sec(2x) + \tan(2x)| - 1$

(b) $2x\cos(2x) + \sin(2x)\ln|\cos(2x)|$

(c) $-\cos(2x)\ln|\sin(2x)| + x\sin(2x)$

(d) $\cos(2x)\ln|\cos(2x)| + 2x\sin(2x)$

(e) $\cos(2x)\ln|\sec(2x) + \tan(2x)| + x\sin(2x)$

**16.** The solution of the initial-value problem

$$xe^y\,dy + \left(\frac{x^2+1}{y}\right)dx = 0, \qquad y(1) = 1,$$

is given by

(a) $3(y-1)e^y + x^2 + 2\ln|x| = 1$

(b) $5(y-1)e^y + x^2 - 2\ln|x| = 1$

(c) $4(y-1)e^y + x^2 + 3\ln|x| = 1$

(d) $(y-1)e^y + x^2 + 2\ln|x| = 1$

(e) $2(y-1)e^y + x^2 + 2\ln|x| = 1$

**17.** If $c_0 = 1$, $c_k = \dfrac{2k-7}{12k}c_{k-1}$, $k \geq 1$, is the recurrence relation corresponding to the indicial root $r = \dfrac{1}{2}$ in the series solution of a differential equation about the regular singular point $x = 0$, then the solution is given by

(a) $y = x^{1/2}\left[1 - \dfrac{5}{18}x + \dfrac{5}{96}x^2 + \cdots\right]$

(b) $y = x^{1/2}\left[1 - \dfrac{5}{12}x - \dfrac{5}{96}x^2 + \cdots\right]$

(c) $y = x^{1/2}\left[1 + \dfrac{5}{12}x + \dfrac{5}{96}x^2 + \cdots\right]$

(d) $y = x^{1/2}\left[1 - \dfrac{5}{12}x + \dfrac{5}{96}x^2 + \cdots\right]$

(e) $y = x^{1/2}\left[1 - \dfrac{5}{12}x + \dfrac{7}{96}x^2 + \cdots\right]$

**18.** A linear homogeneous differential equation with real constant coefficients having the solutions $xe^{2x}$ and $3e^{2x}\sin(2x)$ is given by

(a) $y^{(4)} - 8y^{(3)} + 28y'' - 48y' + 32y = 0$

(b) $y^{(4)} - 8y^{(3)} + 29y'' - 40y' + 16y = 0$

(c) $y^{(4)} + 8y^{(3)} + 29y'' + 52y' + 40y = 0$

(d) $y^{(4)} - 8y^{(3)} + 29y'' - 52y' + 40y = 0$

(e) $y^{(4)} - 6y^{(3)} + 20y'' - 40y' + 32y = 0$

**19.** If $y = \sum_{n=0}^{\infty} c_n x^n$ is a power-series solution of the differential equation $(x^2+1)y'' - 4xy' + 6y = 0$ about the ordinary point $x = 0$, then the constants $c_n$ satisfy the recurrence relation

(a) $c_n = \dfrac{(4+n)(n+5)}{n(n-1)} c_{n-2}, \quad n \geq 2$

(b) $c_n = \dfrac{(2+n)(n-5)}{n(n+1)} c_{n-2}, \quad n \geq 2$

(c) $c_n = \dfrac{(4-n)(n-5)}{n(n-1)} c_{n-2}, \quad n \geq 2$

(d) $c_n = \dfrac{(4+n)(n-5)}{n(n-1)} c_{n-2}, \quad n \geq 2$

(e) $c_n = \dfrac{(4-n)(n+5)}{n(n-1)} c_{n-2}, \quad n \geq 2$

**20.** If the three vectors $v_1 = (1, 2, 3)$, $v_2 = (-1, 0, 4)$, and $v_3 = (3, 2, k)$ of $\mathbb{R}^3$ are linearly dependent, then $k =$

(a) $7$

(b) $-\dfrac{2}{3}$

(c) $-5$

(d) $-2$

(e) $\dfrac{5}{2}$

**21.** If $A = \begin{bmatrix} 2 & 5 \\ 0 & 2 \end{bmatrix}$, then $e^{At} =$

(a) $\begin{bmatrix} e^{2t} & 2te^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(b) $\begin{bmatrix} e^{2t} & 5te^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(c) $\begin{bmatrix} e^{2t} & 5e^{2t} \\ 0 & e^{2t} \end{bmatrix}$

(d) $\begin{bmatrix} e^{2t} & 5te^{2t} \\ 0 & 2e^{2t} \end{bmatrix}$

(e) $\begin{bmatrix} e^{2t} & e^{5t} \\ 0 & e^{2t} \end{bmatrix}$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(e)** | Chain $v_2 = (0,1,0)^T$, $v_1 = (1,0,0)^T$; solutions $v_1$, $v_1 t + v_2$, $\frac{1}{2}v_1 t^2 + v_2 t + v_3$. |
| 2 | **(a)** | Echelon form has rows $(1,-2,3,1)$, $(0,1,-1,2)$, $(0,0,0,k^2-9)$: $k = \pm 3$. |
| 3 | **(d)** | $AK = (0,\ -3\alpha - 2,\ \alpha - 2)^T = -2(\alpha,1,1)^T$ gives $\alpha = 0$. |
| 4 | **(a)** | $\Phi^{-1}F = (-3e^{-t}, 3e^{-3t})^T$, integrate, $X_p = \Phi(3e^{-t}, -e^{-3t})^T = (2,1)^T$. |
| 5 | **(b)** | $F = \int M\,dx + g(y) = x\sin y - y\cos x + \ln\lvert x\rvert + g$, $g = \ln\lvert y\rvert$. |
| 6 | **(c)** | Singular points $2 \pm 3i$; distance from 1 is $\sqrt{1+9} = \sqrt{10}$. |
| 7 | **(e)** | Real/imag parts of $Ke^{(1+i)t}$, $c_1 = 2$, $c_2 = -3$: $X(\pi) = e^{\pi}(-1,-2)^T$. |
| 8 | **(d)** | $p_0 = \frac{1}{2}$, $q_0 = 0$: $r^2 - \frac{1}{2}r = 0$, roots $0, \frac{1}{2}$, sum $\frac{1}{2}$. |
| 9 | **(b)** | $\lambda = -1$: $(-1,1)^T$; $\lambda = 3$: $(1,1)^T$; columns $\mathbf{v}e^{\lambda t}$. |
| 10 | **(e)** | $A\mathbf{v} = \lambda\mathbf{v}$ for each term: $\alpha = 1$, $\beta = -1$, $\gamma = 1$; sum 1. |
| 11 | **(b)** | $A^2 = PD^2P^{-1} = \begin{bmatrix} 13 & -18 \\ 6 & -8 \end{bmatrix}$, entry sum $-7$. |
| 12 | **(a)** | Bernoulli $n = 4$, $v = y^{-3}$: $v' - \frac{6}{x}v = -12x^3$, $\rho = x^{-6}$, $v = 6x^4 + Cx^6$. |
| 13 | **(a)** | Singular at $0, \pm 1$; only $x = 0$ fails ($xP = \frac{1}{x(1-x^2)}$): one irregular point. |
| 14 | **(a)** | $A = 3$, $B = -1$, $C = -2$: $5B + 2A + 5C = -5 + 6 - 10 = -9$. |
| 15 | **(d)** | Variation of parameters, $W = 2$: $u_1 = \ln\lvert\cos 2x\rvert$, $u_2 = 2x$. |
| 16 | **(e)** | Separable: $(y-1)e^y = -\frac{x^2}{2} - \ln\lvert x\rvert + \frac{1}{2}$, times 2. |
| 17 | **(d)** | $c_1 = -\frac{5}{12}$, $c_2 = -\frac{1}{8}c_1 = \frac{5}{96}$, times $x^{1/2}$. |
| 18 | **(a)** | Roots $2, 2, 2 \pm 2i$: $(r^2 - 4r + 4)(r^2 - 4r + 8) = r^4 - 8r^3 + 28r^2 - 48r + 32$. |
| 19 | **(c)** | Coefficient of $x^k$: $(k+2)(k+1)c_{k+2} + (k-2)(k-3)c_k = 0$; shift $n = k+2$. |
| 20 | **(c)** | $\det = 2k + 10 = 0$, so $k = -5$. |
| 21 | **(b)** | $A = 2I + N$, $N^2 = 0$: $e^{At} = e^{2t}(I + Nt)$. |